Counting, Carrying, and Circuits

Week 13 Tuesday — Boolean algebra, and how a computer adds and multiplies

COMSC 2043 • Discrete Structures and Critical Thinking

November 10, 2026

Today's destination

- Notice what **carrying** really is, using the decimal addition you already know.
- See that any set of symbols can be a number system (circles, stars, faces, fish).
- Add and multiply in **binary**, and see it is the same method as the one you learned.
- Meet Boolean algebra and build a **half adder** and a **full adder** from it.
- See why adding 16 bits fast needs a cleverer design: **ripple delay** versus **carry lookahead**.

Warm-up: what happens at $9 + 1$?

Count on your fingers or on paper, out loud if you like:

$$7 + 1 \quad 8 + 1 \quad 9 + 1 \quad 99 + 1$$

- What did you *run out of* at $9 + 1$?
- Where did the extra digit come from?

Counting from 1 to 10 in decimal

0 1 2 3 4 5 6 7 8 9 **10**

- We have **ten** symbols: 0 through 9. Not 10 symbols plus "10".
- At the end of the symbols, we **reset the ones place to 0** and **add 1 to the next place**.
- The "10" is two digits: a 1 in the tens place and a 0 in the ones place.

Add one column at a time: $52 + 53$

place	100s	10s	1s
carry	1		
		5	2
$+$		5	3
	1	0	5

1. **Ones:** $2 + 3 = 5$.
2. **Tens:** $5 + 5 = 10$. Write 0, **carry 1**.
3. **Hundreds:** nothing to add, so the carry drops down: 1.

Do one column, then work sideways to the next.

The same move at $99 + 1$

place	100s	10s	1s
carry	1	1	
		9	9
$+$			1
	1	0	0

1. $9 + 1 = 10$: write 0, carry 1.
2. $9 + 1$ (the carry) $= 10$: write 0, carry 1.
3. Nothing left to add, so the carry becomes a new digit.

Place values: ones $= 10^0$, tens $= 10^1$, hundreds $= 10^2$. **A new column appears every time the last one fills.**

Check: use the carrying rule on bigger sums

$$\begin{array}{rr} & 58 \\ + & 27 \\ \hline & 85 \end{array} \qquad \begin{array}{rr} & 999 \\ + & 1 \\ \hline & 1000 \end{array}$$

Say aloud which columns carried, and what the carry is worth.

Octal: only eight symbols

Digits 0 1 2 3 4 5 6 7. There is no 8, no 9.

0 1 2 3 4 5 6 7 **10** 11

- At $7 + 1$ we run out of symbols, so we write 10 (that number is *eight*).
- Same pattern at the next column: 77 + 1 = 100 (that number is *sixty-four*).
- Place values are powers of 8: 1, 8, 64, 512, ...

Hexadecimal: sixteen symbols

Digits `0 1 2 3 4 5 6 7 8 9 A B C D E F`. We borrow letters because we need six more symbols.

8 9 A B C D E F **10** 11

- `9 + 7 = 10`: sixteen, so a one and a zero. `A + 5 = F`.
- `F + 1 = 10`. `FF + 1 = 100` (that number is two hundred fifty-six).
- Place values: 1, 16, 256, 4096, ...

The pattern, in every base

In base b there are b symbols, the biggest is $b - 1$.

	biggest digit $+1$	**biggest two digits** $+1$
base 10	$9 + 1 = 10$	$99 + 1 = 100$
base 8	$7 + 1 = 10$	$77 + 1 = 100$
base 16	$F + 1 = 10$	$FF + 1 = 100$

"10" always means one group of b and none left over. Only the name of the number changes.

We do not need digits at all

Four symbols: ○ ☆ ☺  (circle, star, face, fish) Agree on an order: ○ < ☆ < ☺ <  .

That is a base-4 system.

- ○ plays the role of 0, ☆ of 1, ☺ of 2,  of 3.
- After  , we are out of symbols: next is ☆○ ("one group of four, none left").

Counting 0 to 8 with circles, stars, faces and fish

decimal	base 4	symbols
0	0	
1	1	
2	2	
3	3	
4	10	
5	11	
6	12	
7	13	
8	20	

Counting 9 to 16

decimal	base 4	symbols
9	21	
10	22	
11	23	
12	30	
13	31	
14	32	
15	33	
16	100	

Row 16 is the ☆◯◯ moment: the same thing as $99 + 1 = 100$.

Add in the invented system



Translate to digits to check: 3 + 1 = 10 in base 4, which is ☆◯. 3 + 3 = 12 in base 4, which is ☆☺.

- Write the answer in the symbols, not the digits. The arithmetic does not care what the marks look like.

Two symbols: binary

Only 0 and 1. Biggest digit is 1.

n	binary
0	0
1	1
2	10
3	11
4	100
5	101
6	110
7	111
8	1000

n	binary
9	1001
10	1010
11	1011
12	1100
13	1101
14	1110
15	1111
16	10000

Place values: $1, 2, 4, 8, 16, \ldots$

The first column fills at 1. So 1 + 1 = 10 (that is two).

Binary addition: the whole rule table

sum	result
$0+0$	0
$0+1$	1
$1+0$	1
$1+1$	10 (write 0, carry 1)

That is the entire addition table. With a carry in, the last case can reach $1+1+1=11$ (write 1, carry 1).

Binary addition: one column at a time

place	16s	8s	4s	2s	1s
carry	1	1	1	0	
		1	0	1	1
+		0	1	1	0
	1	0	0	0	1

1. **1s:** $1 + 0 = 1$.
2. **2s:** $1 + 1 = 10$: write 0, carry 1.
3. **4s:** $0 + 1 + 1 = 10$: write 0, carry 1.
4. **8s:** $1 + 0 + 1 = 10$: write 0, carry 1.
5. **16s:** the carry drops down: 1.

$1011 + 0110 = 10001$, which is $11 + 6 = 17$. Smaller examples: $11 + 1 = 100$ $(3 + 1 = 4)$ and $101 + 011 = 1000$ $(5 + 3 = 8)$.

Check: your turn

Add in binary, then convert both sides back to decimal to check.

`0111 + 0101` `1111 + 0001`

One sentence: **where did each carry come from?**

Binary multiplication: the one-digit table

product	result
0×0	0
0×1	0
1×0	0
1×1	1

A product of single bits is **1 only if both are 1**. Hold that thought: it is AND.

Binary long multiplication: same as the method you know

Decimal: $23 \times 14 = 322$

```
   23
x  14
-----
   92    (23 x 4)
  230    (23 x 10)
-----
  322
```

Binary: $5 \times 3 = 15$

```
   101
x   11
------
   101    (101 x 1)
  1010    (101 x 1, shifted)
------
  1111
```

Each partial product is either the top number or zeros, then shifted. Then **add**, using the rule table.

More binary multiplication

```
   11
x  10
-----
   00
  110
-----
  110
```

$3 \times 2 = 6$

```
    110
x   101
------
    110
   0000
  11000
------
  11110
```

$6 \times 5 = 30$

Where does a computer do this? Boolean algebra

Boolean algebra has two values, true/false, which we write 1/0, and three basic operations:

AND $(A \cdot B)$

A	B	out
0	0	0
0	1	0
1	0	0
1	1	1

OR $(A + B)$

A	B	out
0	0	0
0	1	1
1	0	1
1	1	1

NOT $(\overline{A})$

A	out
0	1
1	0

A trap: Boolean $1 + 1$ is not binary $1 + 1$

In Boolean algebra the plus sign means OR:

$$1 + 1 = 1$$

"true or true is true."

In binary arithmetic the plus sign means add:

$$1 + 1 = 10$$

one, one: write 0, carry 1.

Same marks, different rules. **Always say which system you are in** (Sources, Rules/Assumptions).

A fourth gate: XOR (exclusive or)

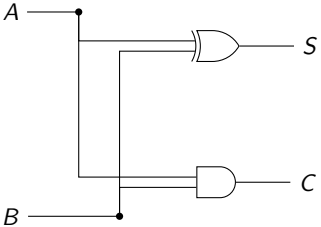
XOR is true when the inputs **differ**. Written $A \oplus B$.

A	B	**XOR**	**binary** $A + B$
0	0	0	00
0	1	1	01
1	0	1	01
1	1	0	10

Look at the last two columns: the **right-hand digit of the sum is the XOR**, and the **left-hand digit (the carry) is the AND**.

The half adder

Adds two bits A and B. Outputs: sum $S = A \oplus B$, carry $C = A \cdot B$.



A	B	C	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

Read CS as a two-bit number: it equals $A + B$.

The problem a half adder cannot solve

Inside a long sum, every column after the first receives three bits: A, B, and the **carry-in** from the column to its right.

```
  1111      <- carries
   01101
+  00111
-------
  10100
```

That is three inputs (A, B, C_{in}) and two outputs (sum, carry-out). We need a **full adder**.

The full adder

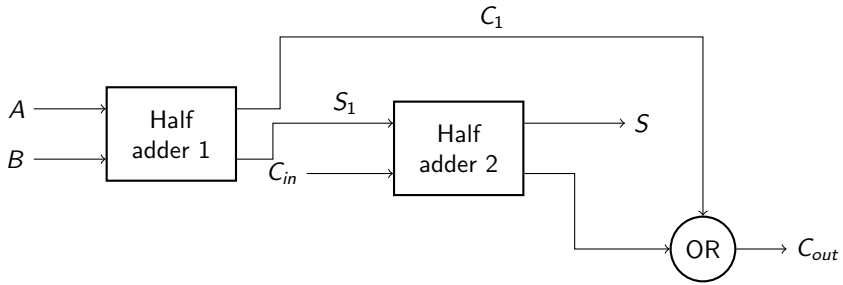
Sum $S = A \oplus B \oplus C_{in}$. Carry-out $C_{out} = A \cdot B + C_{in} \cdot (A \oplus B)$.

A	B	C_{in}	C_{out}	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

Read $C_{out}S$ as a two-bit number: it equals $A + B + C_{in}$ in every row.

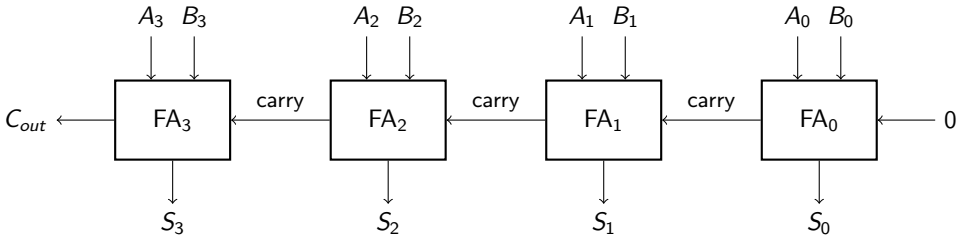
Eight rows is every possible input. **Eight rows is a proof.**

Full adder = two half adders and an OR



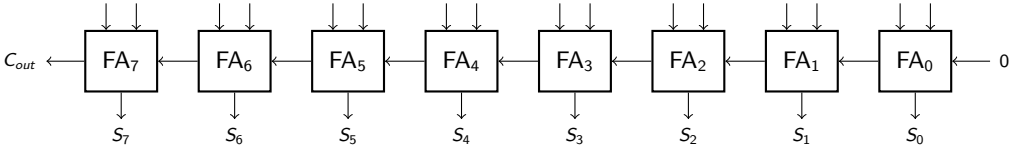
Half adder 1 adds $A + B$. **Half adder 2** adds that sum plus C_{in}. Either one can make a carry, so OR combines them.

Chain them: a 4-bit ripple-carry adder



Each carry ripples left, one stage at a time: the same ripple as $99 + 1 = 100$.

Scale up: an 8-bit ripple-carry adder



Worst case: 11111111 + 00000001 = 100000000 ($255 + 1 = 256$). The carry is created in the *first* column and must travel through **all eight**.

Delay: signals take time

A gate is a physical thing. Its output changes a tiny moment *after* its inputs do. Call that one **gate delay**.

- In one full adder, a carry-in passes through an AND, then an OR: **2 gate delays** per stage.
- The carry into bit i is ready after about $2i + 1$ gate delays; the sum bit needs one more XOR.

bits n	4	8	16	32	64
last sum bit ready after	8	16	32	64	128 gate delays

Delay grows in step with the number of bits: **about** $2n$. The slowest path sets how fast the clock may tick.

Can we get the carries sooner?

Each stage asks one of two questions about its two input bits:

Generate: $G_i = A_i \cdot B_i$
This column makes a carry *by itself* $(1 + 1)$.

Propagate: $P_i = A_i \oplus B_i$
This column *passes along* a carry that arrives (exactly one 1).

Carry out of a column: $C_{i+1} = G_i + P_i \cdot C_i$. Now **substitute** instead of waiting. The carries only depend on the inputs and C_0:

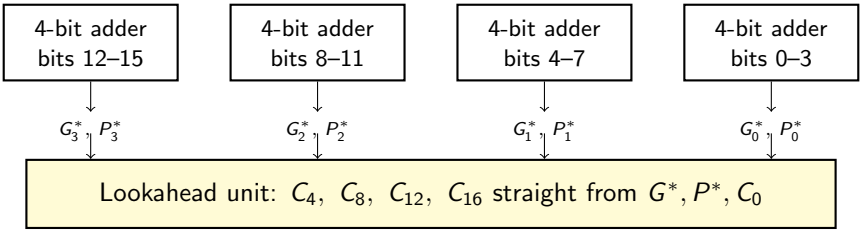
$$C_1 = G_0 + P_0 C_0$$
$$C_2 = G_1 + P_1 G_0 + P_1 P_0 C_0$$

A 4-bit carry-lookahead unit

$$
\begin{aligned}
C_1 &= G_0 + P_0 C_0 \\
C_2 &= G_1 + P_1 G_0 + P_1 P_0 C_0 \\
C_3 &= G_2 + P_2 G_1 + P_2 P_1 G_0 + P_2 P_1 P_0 C_0 \\
C_4 &= G_3 + P_3 G_2 + P_3 P_2 G_1 + P_3 P_2 P_1 G_0 + P_3 P_2 P_1 P_0 C_0
\end{aligned}
$$

Every carry is ready after **3 gate delays** (P and G: 1; the AND terms: 1; the OR: 1). Then sums: $S_i = P_i \oplus C_i$, **4 total**, against 8 for 4-bit ripple.

16 bits: lookahead on top of lookahead



Each 4-bit block reports a **block generate** G^* and **block propagate** P^* ("all four propagate"). The same trick, applied to blocks.

	ripple (16)	**lookahead (16)**
last sum ready after	32	**8** gate delays

Putting multiplication in the circuit

- Each partial product bit is an **AND** of one bit from each number.
- Shifting is just wiring each row one place to the left.
- The rows are added with **full adders**.

So a binary multiplier is AND gates plus the adders we just built.

AI Fluency check

An AI says: "$A + B$ simplifies to A because it worked for $A = 1,\ B = 1$."

- Which system is the plus sign in: Boolean OR or binary addition?
- Is one satisfying row a proof? What would be?

Write one sentence: what does a complete truth table prove that a single example cannot?

Exit ticket and Thursday

Exit ticket (one sentence each):

1. In your own words: what does “carry” mean?
2. Which gate adds one bit, and which gate makes the carry?

Thursday: Show & Tell. **Next Tuesday:** finite-state machines (coin machine, and more) with Mermaid diagrams.